

Patrick Woods

AI and Data Engineer

Hope Mills, NC 28348 | (828) 507-7667 | woods.patrick@icloud.com | [LinkedIn](#) | pnwoods.com | github.com/PNWoods

PROFILE

Technically minded professional with extensive experience in architecting scalable, production-grade intelligent systems that integrate large language models, multimodal pipelines, and retrieval-augmented generation into real-world workflows. Skilled in end-to-end AI solution development, including data ingestion, model orchestration, evaluation pipelines, and human-in-the-loop validation. Proven ability to translate business problems into ML-driven products, balancing automation, performance, and usability.

TECHNICAL PROFICIENCIES

Programming Languages: Python, SQL, TypeScript, Java, C, C++, Swift, Prolog, PowerShell

AI & ML: Retrieval-augmented generation, multi-agent systems, embeddings and hybrid search, LLM quantization and compression, PyTorch, Hugging Face, LangChain, FAISS, Ollama, OpenAI API, Microsoft Copilot Studio, Oracle AI Agent Studio

Data, Backend & Cloud: PostgreSQL, pgvector, Oracle Database, Supabase, REST APIs, Flask, Next.js, Tailwind CSS, AWS (Lambda), Vercel, Cloudflare Tunnel & Zero Trust, Docker, Proxmox, Tailscale, Git, Slurm

Operating Systems: Linux/Unix, Windows, Windows Server, macOS, VMware, Proxmox

High-Performance Computing: Parallel and GPU programming, distributed training on Clemson’s Palmetto cluster

PROFESSIONAL EXPERIENCE

AI and Data Engineer, Fayetteville PWC - Fayetteville, NC **2026 — Present**

- Build AI agents and data infrastructure that give business users fast, accurate, self-service access to their data.
- Built agents in Microsoft Copilot Studio and Oracle AI Agent Studio that automate workflows and surface institutional knowledge.
- Designed and built a centralized data lake consolidating siloed sources into one queryable foundation for analytics and AI.
- Authored complex SQL to power reporting, data validation, and downstream agent workflows.
- Built BI reports and dashboards that turned raw operational data into actionable insights for decision-makers.
- Led internal AI training courses on practical use cases and agent-based tools for business users.

Co-founder & Software Developer, RFP Pilot - Clemson, SC **2025 — Present**

RFP Pilot drafts grounded, compliant RFP responses for government and disaster-recovery responders: every requirement extracted into a compliance checklist, every answer written from the firm’s own past proposals, gaps flagged instead of fabricated.

- Used Retrieval-Augmented Generation to ground AI-generated RFP responses in source documents, reducing hallucinations.
- Built a full document ingestion pipeline for large RFP files (PDF/DOCX), including parsing, chunking, and semantic retrieval.
- Implemented human-in-the-loop review with question-level validation, confidence-based prioritization so reviewers see low-confidence answers first, and traceability linking every answer back to its RFP source sections.

EDUCATION

Bachelor of Science in Computer Science, Minors in AI & Geology | Clemson University | Clemson, SC, 2025
Associate of Applied Science in IT & Operating Systems | Southwestern Community College | Sylva, NC, 2022

UNDERGRADUATE AI RESEARCH – CLEMSON, SC | Spring 2025 – Fall 2025

- Spring 2025: Led high-performance training pipelines on the Palmetto supercomputer for AI experimentation and model convergence, and secured faculty support for RAG research through technical briefings on its deployment benefits.
- Fall 2025: Reduced model footprint and inference latency for edge deployment by applying advanced compression strategies, including low-precision inference and teacher–student learning frameworks.
- Fall 2025: Co-authored “Don’t Waste Bits! Adaptive KV-Cache Quantization for Lightweight On-Device LLMs” (LoViF Workshop, CVPR 2026; arXiv:2604.04722): built the learned per-token precision controller and the base test suite.